Collocation Dynamics in Physics-Informed Neural Networks

Adaptive sampling, diffusion, and solution sharpness

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Why study collocation dynamics?

The physics loss is evaluated on a finite set of points:

$$\mathcal{L}_{\mathrm{PDE}}(\theta; Z) = \frac{1}{N} \sum_{i=1}^{N} r_\theta(z_i)^2, \qquad z_i = (x_i, t_i).$$

- Point locations determine where the PDE is enforced during training.
- Keeping, replacing, or moving points changes coverage and the optimization trajectory, even at the same point budget.
- Concentrating points near difficult features may help, while leaving other regions insufficiently constrained.

What must be evaluated independently?

A small residual at training points does not guarantee a small residual throughout the domain or an accurate solution.

Five ways to place the same interior point budget

Method	Interior point strategy
Fixed Hammersley	Keep the initial low-discrepancy design.
Random resampling	Periodically replace points with fresh uniform random points.
RAD[1]	Sample fresh candidates preferentially where residual magnitude is large.
PACMANN[2]	Move points along the coordinate gradient of squared residual, using Adam ascent.
PACMANN + diffusion	Add domain-scaled Gaussian perturbations to the coordinate-Adam motion.

Residual-based selection versus gradient-based adaptive sampling

RAD selects a new population using residual scores; PACMANN moves the current population using $\nabla_{(x,t)} r_\theta^2$.

Research questions

1. How does the benefit of point-adaptive collocation change across solution-sharpness regimes, for solution and PDE-residual error?
2. Does adding diffusion to PACMANN improve either outcome, and does the effect depend on the regime?

Working hypothesis

Gradient-adaptive methods may offer greater benefits for solutions with sharp features. Both the adaptive-sampling and diffusion benefits remain open questions.

Sharp solution gradients and coordinate gradients of residual scores are different quantities.

Starting problem: viscous Burgers equation

$$\partial_t u + u\,\partial_x u - \nu\,\partial_{xx} u = 0, \qquad (x,t) \in [-1,1] \times [0, 0.99].$$
$$u(x,0) = -\sin(\pi x), \qquad u(-1,t) = u(1,t) = 0.$$

- Viscosity ν controls diffusion in the PDE and the solution regime.
- Train from the PDE and fixed initial/boundary constraints.
- Use numerical reference solutions only for evaluation.

Two distinct parameters

ν changes the physical problem; σ controls stochastic diffusion of collocation points in PACMANN + diffusion.

From the released PACMANN code to a controlled study

Category	Implementation choice
Retained	Four 64-neuron tanh layers, float64, five alternating Adam/L-BFGS stages, upstream sampler settings and coordinate-Adam procedure.
Changed	Match 2,500 interior PDE points from initialization for every method; keep initial/boundary constraints fixed. Use generated references and common final-checkpoint evaluation.
Added / tuned	Gaussian diffusion of point coordinates; select its coefficient separately at each viscosity on development seeds.

- DeepXDE with a PyTorch backend, adapted from the released implementation.
- Evaluate final models and retain all planned seeds, including failures.
- Record actual optimizer work; shared limits do not imply equal computation.

These are disclosed controlled-study choices. Published-result reproduction has not been established.

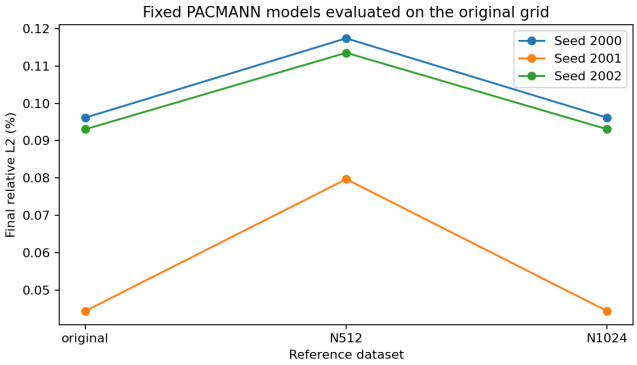
Reference generation and validation at $\nu = 0.01/\pi$

Generated reference

- Chebyshev spatial collocation, degree $N = 1024$.
- Radau time integration.
- Compare on the original 256 spatial points and 100 snapshot times.

Two different checks

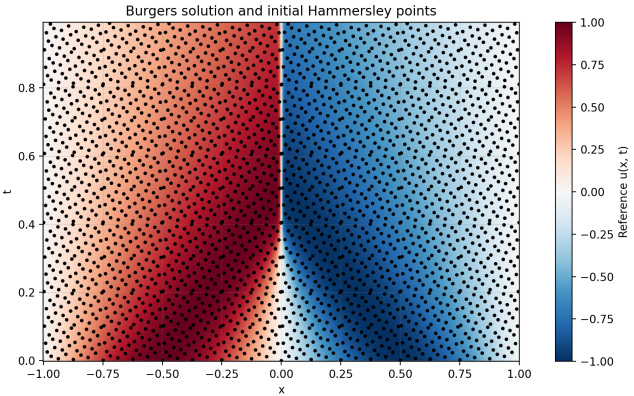
- Reference disagreement: 0.000319% relative L^2.
- N1024 change in fixed-model error scores: less than 0.006% relative.



Each line is one fixed trained model. Only the reference used to score its predictions changes; no retraining occurs.

Generated values are barycentrically interpolated in space at exact shared times. Agreement supports N1024 at this viscosity; it does not certify other viscosities or reproduce published training results.

Matched training and independent evaluation



Training controls

- Paired seeds and initial weights.
- 2,500 PDE; 80 BC; 160 IC.
- Interior updates during Adam only.

Final-model evaluation

- Solution: uniform 1024×199 grid.
- Residual RMS: independent 1024×200 midpoint grid.
- Boundary / initial RMSE.

Initial Hammersley points at $\nu = 0.01/\pi$; paired across methods.

Float64; DeepXDE 1.14.0 / PyTorch 2.7.1; RTX 5070 (12 GB). Ten concurrent seeds per method; method batches run sequentially.

Tune first, freeze settings, then evaluate

Development: three paired seeds

- Seeds 3000–3002.
- Test $\sigma = 0$ and positive values $10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}$.
- Select by mean final solution relative L^2 error.

Evaluation: ten paired seeds

- Freeze sigma before using seeds 4000–4009.
- Use the same sigma for residual-error evaluation.
- Compare all five methods; retain every planned seed.

Frozen diffusion settings: $\nu = 0.01/\pi \Rightarrow \sigma = 10^{-4}$; $\nu = 0.01 \Rightarrow \sigma = 10^{-3}$.

At $\nu = 0.01$, zero is a comparison baseline only: select the best positive sigma and expand a boundary winner under the approved stopping rule. The earlier sweep allowed zero as a candidate. Other samplers retain upstream settings; disclose this tuning asymmetry.

Results: solution and residual accuracy

Results to be added

Complete this comparison after the $\nu = 0.01$ campaign is finished and verified. Include the accepted $\nu = 0.01/\pi$ campaign and the new campaign.

- TODO: Compare all five methods for both error metrics at each viscosity.
- TODO: Show ten-seed means and sample standard deviations; label units.
- TODO: Link each table or figure to its frozen campaign and configuration.

Results: variability and paired comparisons

Analysis to be added

Use the complete seed sets, including unfavorable and failed outcomes.

- TODO: Show individual-seed distributions alongside aggregate results.
- TODO: Assess paired diffusion-minus-PACMANN changes for both metrics.
- TODO: Compare adaptive methods with fixed Hammersley and random resampling.
- TODO: Describe large-error outcomes and the sensitivity of means; keep any trimmed means supplementary.

Analysis: what changes across viscosities?

Interpretation to be added

Connect verified campaign comparisons to the two research questions.

- TODO: Assess whether the effect of diffusion changes across viscosities.
- TODO: Compare method benefits for solution and residual error separately.
- TODO: Discuss variability, reference accuracy, and per-viscosity tuning.

A viscosity-dependent ranking alone does not establish a sharpness effect. Sharpness and feature-region definitions must be independent of adaptive sampling.

Conclusions, limitations, and next steps

- **Evidence:** TODO: State what the completed comparisons support about adaptive collocation and diffusion.
- **Limitations:** TODO: Summarize reference accuracy, seed variability, and the scope of the tested viscosities.
- **Next step:** TODO: State the next bounded study decision.

Interpretation boundary

Distinguish observations from hypotheses about their causes. Improvement in solution error need not imply improvement in independent residual error.

References

[1] C. Wu, M. Zhu, Q. Tan, Y. Kartha, and L. Lu.
A comprehensive study of non-adaptive and residual-based adaptive sampling for physics-informed neural networks.
2023.

[2] C. Visser, A. Heinlein, and B. Giovanardi.
PACMANN: Point adaptive collocation method for artificial neural networks.
2025.

Released-code provenance and controlled-study deviations are recorded in `references/upstream.md`.

Backup: detailed scientific settings

Setting	Frozen configuration
Network / precision	2–64–64–64–64–1, tanh, Glorot normal; float64.
Point budgets	2,500 interior PDE points; 80 boundary and 160 initial points.
Optimizer schedule	Five phases, each with 7,000 Adam steps at 10^{-3}, followed by L-BFGS.
L-BFGS limits	Nominal maxiter 3,000, maxfun 3,750; maxcor 100, gtol 10^{-8}, ftol 0.
Sampling cadence	Every 50 Adam steps; no point updates during L-BFGS.
PACMANN motion	15 coordinate steps at 10^{-5}; $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-7}$ outside the square root.
RAD	100,000 candidates; weight $\lvert r\rvert / \operatorname{mean}(\lvert r\rvert) + 1$; sample without replacement.
Diffusion convention	Per coordinate step, add independent Gaussian noise with componentwise standard deviation $\sigma(L_x, L_t)/\sqrt{15}$.
Evaluation	Uniform solution grid 1024×199 with spatial barycentric reference interpolation; independent residual grid 1024×200.

The retained DeepXDE L-BFGS driver can exceed nominal stage limits. Report recorded iterations, function evaluations, and training time.